

Solutions — IMC Training Sheet, June 10, 2026

Problem 1. (IMC 2010, Day 1, Problem 1) Prove $\int_a^b (x^2 + 1)e^{-x^2} dx \geq e^{-a^2} - e^{-b^2}$.

Solution. For $x > 0$ we have $x^2 + 1 \geq 2x$ by AM-GM, hence

$$\int_a^b (x^2 + 1)e^{-x^2} dx \geq \int_a^b 2x e^{-x^2} dx = \left[-e^{-x^2}\right]_a^b = e^{-a^2} - e^{-b^2}.$$

□

Problem 2. (IMC 2011, Day 2, Problem 3) Compute $\sum_{n=1}^{\infty} \ln(1 + 1/n) \cdot \ln(1 + 1/(2n)) \cdot \ln(1 + 1/(2n + 1))$.

Solution. Set $\varphi(n) = \ln\left(\frac{n+1}{n}\right)$ for $n \geq 1$. Then

$$\varphi(2n) + \varphi(2n + 1) = \ln \frac{2n+1}{2n} + \ln \frac{2n+2}{2n+1} = \ln \frac{n+1}{n} = \varphi(n).$$

Define $g(n) = \sum_{k=n}^{2n-1} \varphi(k)^3$. Using the identity $(u + v)^3 - u^3 - v^3 = 3(u + v)uv$,

$$g(n) - g(n + 1) = \varphi(n)^3 - \varphi(2n)^3 - \varphi(2n + 1)^3 = 3\varphi(n)\varphi(2n)\varphi(2n + 1).$$

Telescoping and noting $g(N) \rightarrow 0$ as $N \rightarrow \infty$ (since $\varphi(k) \leq 1/k$ gives $g(n) \leq n \cdot \varphi(n)^3 \leq 1/n^2$),

$$\sum_{n=1}^{\infty} \varphi(n)\varphi(2n)\varphi(2n + 1) = \frac{1}{3}g(1) = \frac{1}{3}\varphi(1)^3 = \frac{(\ln 2)^3}{3}.$$

□

Problem 3. (IMC 2014, Day 1, Problem 1) Find all pairs (a, b) for which there exists a unique symmetric 2×2 matrix with trace a and determinant b .

Solution. Write $M = \begin{pmatrix} x & z \\ z & y \end{pmatrix}$; the conditions are $x + y = a$ and $xy - z^2 = b$. If (x, y, z) is a solution, so is $(y, x, -z)$; for uniqueness we need $x = y$ and $z = 0$. Then $2x = a$ and $x^2 = b$, so $a^2 = 4b$.

Conversely, if $a^2 = 4b$, then any solution satisfies

$$(x - y)^2 + 4z^2 = (x + y)^2 - 4xy + 4z^2 = a^2 - 4b = 0,$$

forcing $x = y = a/2$ and $z = 0$. So $M = \frac{a}{2}I$ is the unique solution. *Answer:* pairs with $a^2 = 4b$.

□

Problem 4. (IMC 2010, Day 1, Problem 3) Compute $\lim_{n \rightarrow \infty} (x_1 x_2 \cdots x_n) / x_{n+1}$ where $x_1 = \sqrt{5}$, $x_{n+1} = x_n^2 - 2$.

Solution. Let $y_n = x_n^2$. Then $y_1 = 5$ and $y_{n+1} = (y_n - 2)^2$, so $y_{n+1} - 4 = y_n(y_n - 4)$. Inductively $y_n > 5$ for $n \geq 2$, so $y_n \rightarrow \infty$. Now

$$\left(\frac{x_1 x_2 \cdots x_n}{x_{n+1}}\right)^2 = \frac{y_1 y_2 \cdots y_n}{y_{n+1}}.$$

Using $y_{n+1} - 4 = y_n(y_n - 4)$ iteratively,

$$\frac{y_1 \cdots y_n}{y_{n+1}} = \frac{y_{n+1} - 4}{y_{n+1}} \cdot \frac{y_1 \cdots y_{n-1}}{y_n - 4} = \cdots = \frac{y_{n+1} - 4}{y_{n+1}} \cdot \frac{1}{y_1 - 4} = \frac{y_{n+1} - 4}{y_{n+1}} \rightarrow 1.$$

The original ratio is positive, so the limit is 1.

□

Problem 5. (IMC 2011, Day 1, Problem 2) Does there exist a real 3×3 matrix A with $\text{trace}(A) = 0$ and $A^2 + A^t = I$?

Solution. *Answer: NO.* Taking the transpose of $A^2 + A^t = I$ gives $(A^t)^2 + A = I$, so $A = I - (A^t)^2 = I - (I - A^2)^2 = 2A^2 - A^4$, i.e.,

$$A^4 - 2A^2 + A = 0.$$

The polynomial $t(t - 1)(t^2 + t - 1)$ has roots $0, 1, \frac{-1 \pm \sqrt{5}}{2}$, which contain all eigenvalues of A . Eigenvalues of A^2 belong to $\{0, 1, \frac{1 \pm \sqrt{5}}{2}\}$. The conditions $\text{trace}(A) = 0$ and $\text{trace}(A^2) = \text{trace}(I - A^t) = 3$ constrain three eigenvalues $\lambda_1 + \lambda_2 + \lambda_3 = 0$ and $\lambda_1^2 + \lambda_2^2 + \lambda_3^2 = 3$ with $\lambda_i \in \{0, 1, (-1 \pm \sqrt{5})/2\}$. Note $\lambda_i^2 \in \{0, 1, (3 \pm \sqrt{5})/2\}$, and direct

case-checking (the squared values not equal to 1 are irrational and need to cancel in pairs of equal magnitude, etc.) shows no triple satisfies both equations. $\square$

Problem 6. (IMC 2010, Day 2, Problem 3) Subgroup $G \leq S_n$: every $\pi \in G \setminus \{e\}$ has a unique fixed point. Show this fixed point is the same for all π .

Solution. Let G act on $X = \{1, \dots, n\}$. Write $G_x = \{g \in G : g(x) = x\}$ (stabilizer) and Gx (orbit). The hypothesis gives

$$G = \bigcup_{x \in X} G_x, \quad G_x \cap G_y = \{e\} \text{ for } x \neq y.$$

Let $Gx_1, \dots, Gx_k$ be the distinct orbits; since $G_y = gG_xg^{-1}$ when $y = g(x)$, $|G_y| = |G_x|$ is constant on each orbit. By orbit-stabilizer $|Gx_i| = |G|/|G_{x_i}|$. Counting:

$$|G| - 1 = \sum_{i=1}^k \sum_{y \in Gx_i} (|G_y| - 1) = \sum_{i=1}^k |Gx_i| (|G_{x_i}| - 1) = \sum_{i=1}^k |G| \left(1 - \frac{1}{|G_{x_i}|}\right),$$

hence

$$1 - \frac{1}{|G|} = \sum_{i=1}^k \left(1 - \frac{1}{|G_{x_i}|}\right). \quad (*)$$

If two stabilizers $|G_{x_i}|, |G_{x_j}| \geq 2$, the right side of $(*)$ is ≥ 1 , contradicting $1 - 1/|G| < 1$. So at most one stabilizer is non-trivial; the others are $\{e\}$ and contribute zero, forcing the remaining $|G_{x_k}| = |G|$, i.e., $G_{x_k} = G$. $\square$

Problem 7. (IMC 2014, Day 2, Problem 3) Show $|f^{(n)}(x)| < 1/(n+1)$ for $f(x) = \sin x/x$.

Solution. Write

$$\frac{\sin x}{x} = \int_0^1 \cos(xt) dt.$$

Differentiating n times under the integral sign,

$$f^{(n)}(x) = \int_0^1 t^n \cdot g_n(xt) dt,$$

where $g_n(u) \in \{\pm \sin u, \pm \cos u\}$ depending on $n \pmod 4$. Since $|g_n(u)| \leq 1$, with equality only on a measure-zero set,

$$|f^{(n)}(x)| \leq \int_0^1 t^n |g_n(xt)| dt < \int_0^1 t^n dt = \frac{1}{n+1}.$$

$\square$

Problem 8. (IMC 2011, Day 1, Problem 4) Prove $f(t)$ is nondecreasing on $[0, 1]$, where f is the inclusion-exclusion sum.

Solution. Let $\Omega = \bigcup_{i=1}^n A_i$. For $t \in [0, 1]$ construct a random subset $X \subseteq \Omega$ by including each element of Ω independently with probability t . Then for any $C \subseteq \Omega$,

$$\mathbb{P}(C \subseteq X) = t^{|C|}.$$

By inclusion-exclusion,

$$\mathbb{P}(A_1 \subseteq X \text{ or } \dots \text{ or } A_n \subseteq X) = \sum_{k=1}^n \sum_{1 \leq i_1 < \dots < i_k \leq n} (-1)^{k-1} t^{|A_{i_1} \cup \dots \cup A_{i_k}|} = f(t).$$

As t increases, each event $\{A_i \subseteq X\}$ becomes more likely (couple by sharing uniform $U_x \sim \text{Unif}[0, 1]$ for each $x \in \Omega$ and including x iff $U_x \leq t$), so their union too. Hence $f(t)$ is nondecreasing. $\square$

Problem 9. (IMC 2010, Day 1, Problem 5) Given $a, b, c \in [-1, 1]$ with $1 + 2abc \geq a^2 + b^2 + c^2$, prove $1 + 2(abc)^n \geq a^{2n} + b^{2n} + c^{2n}$.

Solution. Consider the symmetric matrix

$$A = \begin{pmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{pmatrix}.$$

Its leading 2×2 minors are $1 - a^2, 1 - b^2, 1 - c^2 \geq 0$, and $\det A = 1 + 2abc - a^2 - b^2 - c^2 \geq 0$. Hence A is positive semidefinite, so $A = B^2$ for some real symmetric B . Let x, y, z be the rows of B . Then $|x| = |y| = |z| = 1$, and $a = x \cdot y, b = y \cdot z, c = z \cdot x$ (where $\cdot$ is the Euclidean inner product).

Consider the n -fold tensor powers $X = x^{\otimes n}, Y = y^{\otimes n}, Z = z^{\otimes n} \in \mathbb{R}^{3^n}$. Then $|X| = |Y| = |Z| = 1$ and $X \cdot Y = a^n, Y \cdot Z = b^n, Z \cdot X = c^n$. The Gram matrix

$$\begin{pmatrix} 1 & a^n & b^n \\ a^n & 1 & c^n \\ b^n & c^n & 1 \end{pmatrix}$$

is positive semidefinite, so its determinant $1 + 2(abc)^n - a^{2n} - b^{2n} - c^{2n} \geq 0$. □

Problem 10. (IMC 2016, Day 2, Problem 10) Prove $\|A^n\| \leq \frac{n}{\ln 2} \|A\|^{n-1}$.

Solution. Set $r = \|A\|$. The matrix norm is submultiplicative, so $\|A^k\| \leq r^k$. Let $\chi(t) = t^n + c_1 t^{n-1} + \dots + c_n$ be the characteristic polynomial. By Vieta's formulas and $|\lambda_i| \leq 1$,

$$|c_k| \leq \sum_{1 \leq i_1 < \dots < i_k \leq n} |\lambda_{i_1} \dots \lambda_{i_k}| \leq \binom{n}{k}.$$

By Cayley-Hamilton, $A^n = -(c_1 A^{n-1} + \dots + c_n I)$, so

$$\|A^n\| \leq \sum_{k=1}^n \binom{n}{k} r^{n-k} = (1+r)^n - r^n.$$

Combined with the trivial bound $\|A^n\| \leq r^n$:

$$\|A^n\| \leq \min(r^n, (1+r)^n - r^n).$$

The two bounds agree when $r^n = (1+r)^n - r^n$, i.e., $(1+1/r)^n = 2$, so $r = r_0 := 1/(2^{1/n} - 1)$. Using $2^{1/n} - 1 = e^{(\ln 2)/n} - 1 > (\ln 2)/n$, we get $r_0 < n/\ln 2$.

Case $r \leq r_0$: $\|A^n\| \leq r^n \leq r_0 \cdot r^{n-1} < \frac{n}{\ln 2} r^{n-1}$.

Case $r > r_0$: The function $h(r) = ((1+r)^n - r^n)/r^{n-1}$ has all-positive numerator of degree $n-1$, so $h(r)$ is decreasing in r (for $r > 0$). Hence

$$\|A^n\| \leq r^{n-1} h(r) \leq r^{n-1} h(r_0) = r^{n-1} \cdot r_0 ((1+1/r_0)^n - 1) = r^{n-1} \cdot r_0 < \frac{n}{\ln 2} r^{n-1}.$$

□